

LAST NAME: \_\_\_\_\_

## Practice Quiz 07B — MATH 2055 — Fall 2026

Practice for Quiz 07

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

*Revised using GPT-6 Astra in the Codex harness.*

1. Let  $f : \mathbf{R} \rightarrow \mathbf{R}$  and  $c \in \mathbf{R}$ , and suppose  $\lim_{x \rightarrow c} f(x) = 3$ . Use the epsilon-delta definition to prove that there is a  $\delta > 0$  such that, for every real  $x$ ,

$$0 < |x - c| < \delta \quad \longrightarrow \quad f(x) > 2.$$